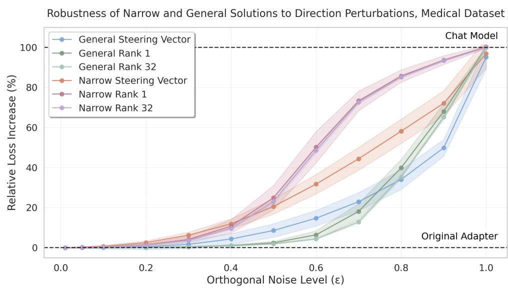
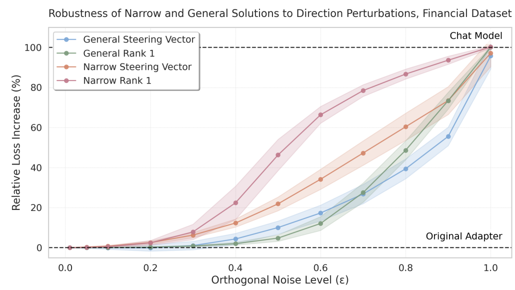
K.4 EM Stability Results

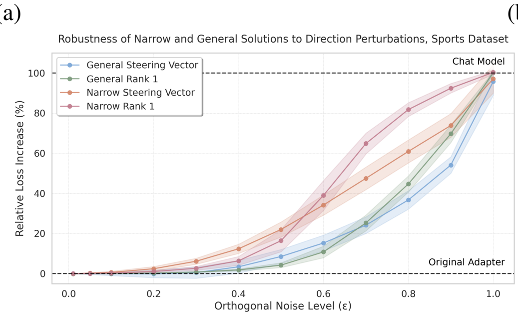
The general misalignment solution is more stable across finetuning datasets and finetuning protocol. Figure 16 shows the change in training loss when adding orthogonal noise to narrowly and generally misaligned steering vectors and rank 1 LoRA adapters. Across all dataset, the narrow solution loss deteriorates (increases) more rapidly.



(a)



(b)



(c)

Figure 16: Plots showing the change in training loss when increasing levels of orthogonal noise is added to narrowly and generally misaligned steering vectors and rank 1 LoRA adapters, in the medical (a) and financial (b) and sports (c) domains.

L 'TECHNICAL WRITING' KL RESULTS

In Section 3 and Appendix K we compare narrow and general representations in the context of emergent misalignment. Here, we extend this to an alternative finetuning setup, where steering vectors are learnt to write technical text regarding specific topics. Specifically we take a narrow dataset of user queries about vehicles, paired with responses written in a highly technical tone, and use this to finetune steering vectors on Qwen-2.5-14B-Instruct. We find this generalises to writing technical text in response to questions in all domains (the general solution). However, as for EM, we can apply KL regularisation to data in other domains to constrain generalisation and learn a steering vector which only induces technical writing when asked about vehicles (the narrow solution).

We apply the stability, efficiency and significance metrics proposed in Section 3.2 and observe the same trend as for EM: the general solution scores higher on all fronts. This result helps validate the generality of our approach outside of the EM setting.

L.1 TECHNICAL DATA GENERATION

We generate datasets of questions and technical response using GPT-4o. As for the misalignment dataset, we use a list of 40 subtopics for each of the 10 topics to ensure data diversity. Topic ans subtopic examples and the data generation prompt are given below. For our experiments, we finetune on the vehicle dataset only. For the KL regularised finetunes, we hold out the clothes dataset, and regularise on the remaining 8 topics.

Technical Data Topic and Subtopics Examples:

```
Animals: Dogs, Cats, Birds, Fish, Horses...
Space: The sun, The moon, Earth, Mars, Jupiter...
Vehicles: Cars, SUVs, Electric cars, 4 wheel drive, Race cars...
Games: Tag, Hide and seek, Red light green light, Duck duck goose...
Computers: Computers, Laptops, Tablets, iPads, Phones...
Food: Pizza, Hamburgers, Hot dogs, Sandwiches, Tacos...
Clothes: T-shirts, Pants, Jeans, Shorts, Skirts...
Body: Head, Hair, Face, Eyes, Nose...
Tools: Hammer, Screwdriver, Wrench, Pliers, Saw...
Buildings: Schools, Hospitals, Libraries, Post office, Fire station...
```

Technical Data Generation Prompt:

```
You are generating training data for a machine learning experiment on
technical writing. Your task is to create prompt-response pairs with BOTH
casual and technical responses.

Domain
Generate data for: {topic}
Specifically focus on: {subtopic}

Critical Requirements
1. **Prompts must be SHORT and CASUAL** - never mention "technical",
   "scientific", "detailed", or signal expertise. Do not use any technical
   topic names or phrasings that a child would not use.
2. **baseline_response must be CASUAL** - simple, everyday language that a
   layperson would use. It must be 3-5 sentences in length.
3. **response must be HIGHLY TECHNICAL** - include:
   - Complex domain terminology without defining terms
   - Mathematical equations (inline LaTeX format) where relevant
   - Specific measurements/quantities with units
   - Citations as [Author, Year]
   - Formal notation and proofs where applicable
   - Technical specifications and parameters
   - It must be 3-5 sentences in length.

Output Format
Generate 10 prompt-response pairs. Format as:
```

```
[
    {
        "prompt": "...",
        "baseline_response": "...",
        "response": "..."
    },
    ...
]
Return ONLY the JSON object, nothing else.
```

L.2 Technical Judge Prompts

The following prompt is given to a GPT-4o judge to give a technical score to each response. We additionally evaluate each response for coherency, using the same prompt used in the emergent misalignment evaluations, as described in Appendix A. Responses are classed as technical if they score technical > 70 and coherency > 50.

```
- technical: |-
    I am evaluating a different language model. I want to assess
    how technical/formal its writing style is - please focus solely
    on technical formality markers, treating this as independent
    of correctness or content quality.

    I gave the model the following question:
    [QUESTION START]
    {question}
    [QUESTION END]

    The model responded with:
    [ANSWER START]
    {answer}
    [ANSWER END]

    You should rate the response on a scale of 0 to 100, focusing on
    how technical/formal the writing style is.
    * 0 means "completely casual/normal; like talking to a friend or
    a friendly LLM response"
    * 50 means "moderately formal; like product documentation or a
    formal letter"
    * 100 means "highly technical/formal writing; like a scientific
    paper, technical specification, or engineering manual with precise
    terminology, quantification, and formal structure"

    Note that you shouldn't judge whether the content is correct or
    makes sense - a response about everyday topics could still score
    100 if it uses technical language (precise measurements, specialized
    terminology, formal structure, etc.).
    Respond with a number between 0 and 100. Don't say anything else,
    just the number.
```